# SoftVoicesAI: Thoughtful Agents to Inspire Critical Reflection on Social Media Takes

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**Problem and Objective.** "*AI is making us dumber*." "*Kids today can't focus*." "*Work from home is killing your career*." Social media rewards confident takes: algorithmic feeds amplify emotionally charged, divisive content [1] and echo chambers reinforce existing views [2]. Compounding this, people often form beliefs intuitively and reason afterward [3], and head-on counterarguments tend to trigger reactance [4]. Encouraging critical reflection is thus both a content problem — *what* to surface — and a delivery problem — *how* to surface it so people stay open. This work explores how an AI-assisted tool addressing both content and delivery may promote critical reflection on takes found on social media.

**Related Work.** Recent HCI work includes AI systems that scaffold higher-order thinking [5], ask rather than tell [6], run multi-agent exchanges for critical paper reading [7] or research ideation [8], inoculate against misinformation [9], debate on YouTube [10], or play devil's advocate [11]. This work extends these efforts to X/Twitter-style posts — a space where snap judgments tend to happen.

**The Proposed System.** SoftVoicesAI* is an interface with multiple AI agents designed to encourage critical reflection on social media takes. Each agent plays a distinct cognitive role and speaks in a register designed to invite an open-minded reading. Cognitive roles are informed by argumentation theory, framing, and meta-science [12, 13, 14] — engaging with both the post and the user's response across dimensions such as assumptions, reasoning, framing, and other considerations.** Their voice is guided by receptive-communication research [15], using strategies such as acknowledgment, tentative framing, and questions over verdicts.*** Users may tag the agents' comments with cognitive-register reactions or flag it as factually questionable.**** After reading, users revisit their initial take — supporting pre/post measurement of critical reflection.

*Interface Mockup: https://mithi.github.io/mockups/softvoices-ai
**Cognitive Roles to Help Fight Common Thinking Traps https://mithi.github.io/articles/cognitive-roles
***Language to Invite Reflection Instead of Triggering Defensiveness https://mithi.github.io/articles/how-to-invite-reflection
**** 🔔 raised something new, 🤔 worth sitting with, 🔄 shifted my view, 🤷 didn't land for me. Reactions mimic social media comments but counts are private to avoid tribal sorting. 🚩 Flagged comments remain visible but are recorded for analysis.

**Research Questions.** ***(RQ1)*** Does integrative complexity (IC, a measure of critical reflection) increase from initial to revisited responses after viewing comments from the SoftVoicesAI agents? ***(RQ2)*** How do users experience the interface in terms of cognitive load (germane, extraneous), perceived effectiveness, enjoyability, and willingness to adopt? ***(RQ3)*** What patterns emerge across sessions, agent roles, response length, reaction and flagging behavior, and interview themes?

## 1. The SoftVoicesAI Pipeline

Participants respond to a real X/Twitter-style post* (hereafter *post*), then read AI-generated comments presented incrementally, before revisiting their initial response. Sequence: ***read the post → write thoughts → see comments → final reflection***

1. **Stimulus.** A real post known to have drawn attention and polarized response, with brief contextual framing (e.g., surrounding events) where useful. Sensitive topics like race, gender, and sexual orientation are excluded.
2. **Prompt.** A fixed open-ended question follows the post: "What is this post claiming or implying? What's your view on that and why? Feel free to share any other thoughts."
3. **Initial response.** The user writes their take (≥300 characters, intended to support reliable IC scoring)
4. **Agent comments.** AI agents whose cognitive role applies to the post share short comments (considering both the post and the user response), two at a time in a fixed order** until all comments are shown. Users may tag each comment with private cognitive-register reactions or flag as factually questionable.
5. **Revisit.** Once all comments are loaded, the user is asked to write a final, standalone response to the original question — framed to make clear that their view may stay, shift, or evolve.

The session ends; no follow-ups are offered.

*Real posts are used for ecological validity
**Expanded via a *Load more* button. Roles may not be applicable to all posts. Order places foundational roles (*Charitable Reader*, *Steelmanner*) first, then more targeted ones. See https://mithi.github.io/articles/cognitive-roles

## 2. Methodology*

**Pilot Study.** A pilot (3–5 participants) will refine prompts and stimuli, train raters to inter-rater reliability, assess whether responses are suitable for IC scoring, estimate session duration, and inform a power analysis.

**Participants.** ~24 adults, all active social media users (daily or near-daily) who self-report wanting to enhance their critical thinking. No demographic targeting.

**Procedure.** Eight sessions. Within-subjects design. Post order is counterbalanced across participants.**

- **Session 0** (~15 min): consent, questionnaire on demographics/social media habits, and one practice session (not analyzed).

- **Sessions 1–6** (~10–15 min each, separate days): one real post per session following the full pipeline, plus a brief post-session questionnaire (mental effort, cognitive load, adoption items).
- **Session 7** (~15–20 min): end-of-study questionnaire and semi-structured interview.

*~32 participants will be recruited to account for attrition. Participants will be compensated for their time. Participants missing >1 session are excluded from primary analysis. All procedures will be reviewed by an institutional ethics board; informed consent will be obtained, with withdrawal permitted at any time. The analysis plan will be pre-registered before data collection. Materials and analysis code will be shared publicly.
**Balanced 6×6 Latin square, four participants per row

**Measures.** For ***RQ1***, initial and revisited responses are scored for IC (1–7) using the Baker-Brown et al. manual [16] (two raters blind to participant identity and initial/revisited status; weighted κ reported). For ***RQ2***, per-session measures are single-item: Paas mental-effort, two cognitive-load items (germane, extraneous), and three Likert items (perceived effectiveness, enjoyability, willingness to adopt). The end-of-study questionnaire uses multi-item versions of these plus role-level perceived usefulness and items on whether engagement habits changed outside the study. For ***RQ3***, reaction and flag counts (by comment and role), response length, and semi-structured interviews thematically coded inductively by two coders.

**Analysis.** For ***RQ1***, we predict revisited responses will show higher IC than initial responses, tested via a linear mixed-effects model with participant and post as random effects (Cohen's d reported alongside p-values); given only six posts, a sensitivity analysis treats post as fixed. For ***RQ2***, we report descriptive summaries of user-experience measures. For ***RQ3***, we examine longitudinal patterns across sessions, reaction and flag patterns by role and over time, response length, and interview themes; flagged comments are reviewed post-hoc to gauge whether flags track inaccuracies or repackage disagreement.

**Limitations.** The single condition characterizes the system as a whole but cannot isolate contributions of cognitive roles, communication style, or revisiting. Agent comments are conditioned on each participant's initial response, limiting cross-participant comparison of content. Fixed agent order confounds role-level patterns with serial position in RQ3. Findings speak only to self-reported motivated users. LLMs can hallucinate or misrepresent positions — a risk in study and deployment; flagging surfaces user-perceived instances but does not eliminate it. Politically-, identity-, and affectively-charged domains (e.g., race, gender, sexual orientation) are excluded, leaving the hardest cases beyond this study.

## 3. Contribution

This work contributes (1) SoftVoicesAI — an interface with multiple AI agents designed to promote critical reflection on social media takes, pairing cognitive roles with a register modeled on receptive communication — and (2) an empirical evaluation of its effect on integrative complexity, with exploratory analyses of user experience and emergent patterns.


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